Malith Jayasinghe

Software Engineer | Backend & AI Engineering

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PROFESSIONAL SUMMARY

Software Engineer with 2 years of experience across industry roles and freelance projects, building full-stack applications and AI-powered automation systems. Experienced in backend development, API integrations, and LLM-based solutions, with hands-on experience using AI-assisted development tools to improve development workflows, code quality, and productivity. Interested in building reliable and scalable software products.

WORK EXPERIENCE

Junior AI Automation Engineer - Aahaas 06/2026 – Present | Colombo

- Developed Travel Quote GPT, an AI-powered travel quotation and itinerary generation solution using n8n, JavaScript, OpenAI, Gemini, APIs.
- Automated travel quotation processing, reducing preparation time through intelligent workflow automation.
- Developed and integrated backend APIs across multiple systems, implementing data synchronization, validation, and prompt-injection safeguards to improve workflow reliability.
- Optimized database queries and LLM token usage, improving database performance and reducing AI processing costs.
- Maintained and enhanced production AI automation systems while evaluating LLM solutions.

Intern Software Engineer - One Billion Tech 05/2025 – 05/2026 | Colombo

Million Spaces **- Workspace Booking Platform**

Technologies: Spring Boot (Microservices), React, AWS, MongoDB

- Developed and maintained Spring Boot REST APIs and responsive React interfaces for a workspace booking platform.
- Implemented server-side caching in Spring Boot to optimize backend performance and response times.
- Integrated and optimized the Google Calendar API to improve booking synchronization and scheduling reliability.
- Collaborated with the development team to migrate backend services to AWS serverless architecture, improving scalability and reducing infrastructure costs.
- Co-developed Spacey, an AI-powered chatbot providing automated assistance to platform users.

1HR - Human Resource Management System

Technologies: Node.js, React, PostgreSQL, AWS, Azure, Terraform

- Developed full-stack features for a cloud-native HR management system covering employee management, attendance, and financial reimbursements.
- Worked on multi-cloud infrastructure using Terraform.
- Developed serverless application components to improve scalability and reduce operational overhead.

PROJECTS

AI Agent System for E-commerce (personal project) 07/2026 – Present

Technologies: Python, LangGraph, FastAPI, MCP, PostgreSQL, pgvector, Redis, Gemini API, Ollama, Next.js, TypeScript

- Developing a multi-agent AI platform for e-commerce, featuring intelligent customer support, shopping assistance and dynamic agent routing.
- Implemented an MCP server enabling secure, customer-specific access to e-commerce catalogue data for external AI assistants.
- Designed secure AI workflows with prompt-injection protection, data isolation, response validation and human handover mechanisms.
- Developed RAG-based product search with support for both cloud-based and locally hosted language models.
- Built a practice mode where shop owners grade AI replies before the AI goes live.

Sri Lankan Legal Information Platform (personal project) 02/2026 – Present

Technologies: NextJs, Python, RAG, Gemini API, LangChain

- Developed a legal information platform focused on helping users explore and understand Sri Lankan legislation.
- Implemented an AI-powered legal assistant that retrieves relevant legal information and provides source-referenced responses.
- Developed features including a legal library, law firm directory, blog and legal forms.

JobCeylon - AI Powered Job Platform 07/2026 – Present

Technologies: Next.js, TypeScript, Supabase, GitHub Actions, Vercel

- Contributed to a live job platform that aggregates Sri Lankan job listings, matches them to users, and sends Telegram/email alerts.
- Implemented subscription tier limits and pricing logic, and fixed PayPal payment flow issues.
- Improved UX with job filters, a simpler application flow, and mobile-responsive layouts.

TECHNICAL SKILLS

AI & LLM
LangGraph, LangChain, RAG, OpenAI, Gemini, Ollama APIs

Programming Languages
Python, Java, JavaScript, TypeScript, PHP, C#

Database Management
MySQL, PostgreSQL, MongoDB, Redis

Development Tools
Claude Code, Cursor, GitHub Copilot, Git, GitHub, GitHub Actions, Postman

Backend Technologies
Spring Boot, Node.js, REST APIs, Microservices

Frontend Development
React, Next.js, Redux, HTML5, CSS3, Tailwind CSS

Cloud & DevOps
AWS (Lambda, EC2, S3, ECS, SQS, RDS), Terraform, Docker, CI/CD

EDUCATION

BSc (Hons) in Information Technology 2021 – 2025 | Malabe
Specialization in Software Engineering
Sri Lanka Institute of Information Technology (SLIIT)

NVQ Level 4 in Information Technology 2020 – 2021
Technical College, Warakapola

GCE Advanced Level - Physical Science Stream 2020
Maliyadeva College

CERTIFICATES

- AWS Educate Getting Started with Serverless — AWS | 2026
- AWS SimuLearn: Cloud Practitioner — AWS | 2025
- Foundational C# with Microsoft — freeCodeCamp | 2024
- Front End Development Libraries — freeCodeCamp | 2024

REFERENCES

Mr. Anjana Dasanayake, *Cloud Operations Engineer,* Citecore
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Mr. Dilshan Amarasinghe, *Software Engineer,* 1 Billion Tech
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